

# Connor Lindsell

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## PROFESSIONAL SUMMARY

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Final-year Mechanical and Mechatronic Engineering student with design ownership of live capital projects in an aluminium extrusion plant, working from concept and CAD through FEA verification, GD&T manufacturing drawings, supplier engagement and on-site commissioning. Capstone research applies machine learning to robotic manipulation for adaptive industrial packing and assembly, connecting automation research to real manufacturing constraints. Experienced delivering to production schedules and site constraints, where designs are judged on installed performance rather than on paper.

## EDUCATION

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**Bachelor of Engineering (Honours), Mechanical and Mechatronic**

**Feb 2022 – Nov 2026**

University of Technology Sydney (UTS)

- **Distinction Average;** GPA: 6.03, WAM:77.75
- **Active Research (Capstone):** Learning Based Robotic Manipulation for Adaptive Industrial Packing and Assembly

## TECHNICAL SKILLS

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**Design & Simulation:** Autodesk Inventor, Inventor Nastran, SolidWorks, Fusion 360, FEA (Static Structural), Topology Optimisation (SolidWorks & Inventor), ANSYS Mechanical (Static Structural, Steady State Thermal) ANSYS Fluent (CFD)

**Robotics & AI:** ROS 2 & ROS 1, Gazebo (Ignition), RViz, OpenCV, RGB & Depth Cameras, RL, LfD, Hierarchical Control System

**Mechanical & Manufacturing:** GD&T, DFM / DFA, Sheet Metal Design, 3D Printing, Laser Cutting, Orca Slicer, Creality Print

**Tools:** GitHub, VS Code, MATLAB, Arduino (Uno, Mega, ESP32), Microsoft 365 (Teams, Word, Excel, OneDrive)

**Languages:** C++, C, Python

## TECHNICAL EXPERIENCE

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**Undergraduate Engineer (Mechanical & Mechatronic)**

**Dec 2025 – Present**

Capral Aluminium, Smithfield, NSW

- Developed Mechanical and Mechatronic Systems through design and automation projects in a live aluminium extrusion environment, managing work from concept through implementation using data-driven design decisions.

Hot Line Billet Roller System Upgrade

- Analysed production and maintenance data to define a billet-handling improvement opportunity associated with a 1.17% scrap rate, 69.95 tonnes of billet scrap and 42.53 hours of project-relevant downtime in 2025.
- Redesigned the roller and transfer system by reducing roller centres 41.2% and the maximum transfer pitch 42.3%, increasing billet support by 70% and enabling stable handling of billets down to 228 mm.
- Simplifying the overall mechanism by eliminating unreliable anvil mechanisms and hydraulic cylinders with pneumatic cylinders resulting not only a mechanical improvement but also improvement in safety.
- Utilised Autodesk Inventor to create and design GD&T-compliant drawings from 3D models and validated through FEA simulations including static structural and steady state thermal analyses in Inventor Nastran.

Packing Line Extension and Automation Upgrade

- Designed and engineered mechanical assemblies using Autodesk Inventor, applying FEA methods to validate structural mechanisms, delivering a projected 33% increase in total line productivity.
- Performed load calculations for motor selection, pneumatic sizing and bearing validation to ensure design compliance.
- Produced GD&T-compliant manufacture ready drawings, negotiated with suppliers and completed QA inspections to ensure smooth manufacturing and delivery, whilst managing a team of 8 during the installation process to ensure the installation matches intended design.

TECHNICAL PROJECTS

WordleBot – Autonomous Wordle Playing Robot Mar 2026 – May 2026

Robotic Studio 2 | UTS

- Designed and trained a MaskablePPO reinforcement-learning policy to sequence pick-and-place operations across a Wordle game board, using action masking, curriculum learning and reward shaping to achieve a 100% success rate across 20 evaluation episodes.
- Outperformed a deterministic nearest-neighbour baseline by reducing average robot travel distance by 5%, demonstrating globally optimised task sequencing rather than locally greedy motion selection.
- Developed and integrated the ROS 2 and MoveIt 2 low-level control architecture for a UR3e and OnRobot RG2 gripper, implementing collision-aware motion planning, attached-object handling, trajectory scoring, collision recovery and mission control.

Pliers Design Challenge for Additive Manufacturing Apr 2025 – May 2025

Additive Manufacturing | UTS

- Applied topology optimisation in SolidWorks, reducing effective weight by 22% (35.2 g to 27.6 g) while increasing load capacity by 82% (64.6 N to 117.5 N) through iterative prototyping and testing of six designs.
- Refined optimised meshes in Blender for printability, then tuned Creality Print 6 slicer settings, including infill patterns, support strategies and print parameters, to balance weight reduction with functional strength.
- Conducted experimental testing using a force meter to measure load at failure, documented fracture locations and validated optimisation outcomes under representative mechanical loading.

PROFESSIONAL EXPERIENCE

Disability Support Worker Apr 2025 – Feb 2026

What Ability, Sydney Metropolitan and Central Coast

- Supported neurodiverse clients through tailored activity planning and real-time adaptive problem solving in dynamic, unscripted environments.
- Demonstrated independent decision-making, emotional intelligence, and calm leadership in unpredictable conditions.

Office Manager Mar 2025 – Oct 2025

Essential Health Chiropractic, North Kellyville, NSW

- Implemented process improvements and software automation to enhance operational efficiency consistency.
- Coordinated team operations and stakeholder communication, strengthening accountability and service performance.

VOLUNTEERING

Project Protect July 2024 – Jan 2025

Ski Racing Australia (SRA)

- Collaborate with a multidisciplinary team to research, design and prototype alternative safety solutions.
- Engage with stakeholders, industry professionals and academics to make evidence-based design decisions.

TRANSFERABLE SKILLS

**Problem Solving:** Analyse and resolve complex technical issues under pressure while maintaining operational continuity.

**Rapid Learning:** Rapidly learn new tools, technologies and systems, applying new knowledge to improve engineering performance and workflow.

**Leadership:** Led cross-functional teams of up to 50 staff, applying project coordination, conflict resolution and time-management skills to maintain performance and motivate team members.

REFERENCES

Available upon request